📄 DOCUMENT 6

Core Technology Architecture & Enterprise Integration Blueprint

Hawassa MESOB One-Stop Service Center

Prepared For:
H.E. Mayor Tiratu Beyene
Hawassa City Administration

Prepared By:
Chief Enterprise Architect & Lead Consultant
Hawassa MESOB One-Stop Service Program

PART 1

(Sections 1–5: Enterprise Architecture Principles, Logical Architecture Layers, API Endpoint Architecture, Data Exchange Mesh Alignment, Localized Integration Model)

---

1. ENTERPRISE ARCHITECTURE PRINCIPLES

The Hawassa MESOB system must not be a collection of disconnected applications.

It must operate as a structured Enterprise Architecture (EA) aligned with:

- National MESOB API Gateway/Mesh model
- Federal digital strategy
- City-level governance needs
- Future Phase 2 expansion

1.1 Architectural Goals

The architecture must ensure:

1. Modular service expansion
2. API-first integration
3. Data ownership by Hawassa City
4. Vendor-neutral scalability
5. Secure inter-bureau communication
6. Real-time executive visibility

1.2 Architectural Constraints

The following realities must be considered:

- Some bureaus lack IT systems
- Ethio Telecom may provide infrastructure & operational platform
- API responsibilities currently undefined
- National mesh architecture must be respected
- Time-to-launch is politically sensitive

The architecture must solve these constraints structurally.

---

2. LOGICAL ENTERPRISE ARCHITECTURE LAYERS

The MESOB system will follow a six-layer logical architecture.

2.1 Layer 1 – Presentation Layer

This includes:

- MESOB service counters
- Citizen-facing portal (future enhancement)
- Queue management displays
- Executive dashboards

Responsibilities:

- User interaction
- Service submission
- Status tracking
- Ticket issuance

No business logic should reside here.

2.2 Layer 2 – Application / Workflow Layer

This is the service orchestration engine.

It handles:

- Service configuration (47 Phase 1 services)
- SLA timers
- Routing rules
- Approval sequencing
- Notification logic
- Document handling

This layer must be centrally managed.

2.3 Layer 3 – Integration Layer (Critical Layer)

This includes:

- API Gateway
- Authentication middleware
- Message routing engine
- Error-handling layer
- Logging & monitoring

All bureau communication must pass through this layer.

Direct database-to-database access is prohibited.

2.4 Layer 4 – Bureau Backend Systems

These are:

- Land registry database
- Revenue tax system
- Trade licensing system
- Investment registry
- Health & education systems

For Category B bureaus:

New backend systems must be built here.

2.5 Layer 5 – Data Layer

Includes:

- Transaction database
- Payment ledger
- Document storage
- Master data registry
- Analytics warehouse

2.6 Layer 6 – Infrastructure Layer

Includes:

- Hosting servers
- Network switches
- Firewalls
- Storage arrays
- Backup systems
- Connectivity to national mesh

---

3. API ENDPOINT ARCHITECTURE DESIGN

The API architecture must follow strict discipline.

3.1 API Gateway Model

All external and internal APIs must pass through:

Central API Gateway

Functions:

- Authentication
- Authorization
- Rate limiting
- Traffic control
- Logging
- Version control
- Error handling

The gateway acts as the single entry point.

3.2 API Classification Model

APIs are divided into:

1. Read APIs

Example:

- Retrieve lease record
- Retrieve tax status
- Retrieve business registration

2. Write APIs

Example:

- Update ownership
- Post payment
- Issue license

3. Event APIs

Example:

- Payment confirmation event

- Approval completion event
- SLA breach event

3.3 API Standard Requirements

All APIs must:

- Use REST architecture
- Use JSON format
- Use HTTPS encryption
- Require token authentication
- Provide structured error codes
- Maintain audit logs

3.4 Version Control Requirement

Each API must include:

- Version number (v1, v2, etc.)
- Change log
- Backward compatibility policy

No API update allowed without Technical Steering Committee approval.

---

4. DATA EXCHANGE MESH ALIGNMENT

(National Integration Context)

At national level:

MESOB taskforce operates API Gateway/Mesh architecture.

Hawassa MESOB must integrate as a node within this mesh.

4.1 Local-to-National Integration Flow

Example:

Citizen → Hawassa MESOB → Local API Gateway → National Gateway → Federal System

Possible federal integrations:

- National ID verification
- Federal tax cross-check
- Land dispute database (if applicable)

4.2 Local Autonomy Preservation

Important:

Hawassa MESOB must retain:

- Local data control
- Local service configuration authority
- Independent reporting capability

National integration should not override city autonomy.

4.3 Mesh Compliance Requirements

Hawassa must:

- Use approved API security protocols
- Follow federal data standards
- Register APIs in national directory
- Maintain interoperability compliance

---

5. LOCALIZED INTEGRATION MODEL

(Solving the API Responsibility Problem)

This is the most critical architecture section.

5.1 Integration Responsibility Layers

We define three integration layers:

Layer A – Core MESOB Platform
Layer B – API Gateway
Layer C – Bureau Backend Systems

Responsibility must be assigned clearly per layer.

5.2 Recommended Responsibility Allocation

Given current context:

Ethio Telecom likely provides:

- Infrastructure hosting
- Possibly core platform
- Network connectivity
- Queue management

But:

Responsibility for bureau backend and APIs must be explicitly contracted.

5.3 Proposed Enterprise Integration Governance Model

1. MESOB Integration Authority (within City)
    - Owns API standards
    - Approves integration designs
    - Controls documentation
2. Vendor Integration Team
    - Builds API gateway
    - Develops backend systems for Category B bureaus
    - Wraps APIs for Category A bureaus
3. Bureau Technical Focal Points
    - Validate data mapping
    - Test endpoints
    - Approve go-live

5.4 Enterprise Architecture Control Board

Must be created before implementation.

Responsibilities:

- Approve system design
- Approve API schemas
- Review security compliance
- Validate scalability planning

Without this board, vendor-led architecture may become fragmented.

ENTERPRISE ARCHITECTURE SUMMARY (PART 1)

The Hawassa MESOB architecture must:

✓ Be layered
✓ Be API-first
✓ Align with national mesh
✓ Protect city data ownership
✓ Enforce integration discipline
✓ Clearly assign API responsibilities

Failure to define these clearly will cause:

- Vendor lock-in
- Integration delays
- Political risk
- Revenue reconciliation errors

END OF DOCUMENT 6 – PART 1

Covered:

✓ Enterprise architecture principles
✓ Logical architecture layers

✓ API design standards
✓ Data exchange mesh alignment
✓ Integration governance model

Reply with:

“Proceed with Document 6, Part 2.”

Next we will produce:

✓ Detailed enterprise architecture diagram (textual structural format)
✓ Localized data center requirements
✓ Hosting topology options
✓ Security zoning diagram logic
✓ Scalability & future-proofing model.

📄 DOCUMENT 6

Core Technology Architecture & Enterprise Integration Blueprint

Hawassa MESOB One-Stop Service Center

PART 2
(Sections 6–10: Enterprise Architecture Diagram Structure, Hosting Topology, Localized Data Center Requirements, Security Zoning Model, Scalability & Future-Proofing)

---

6. ENTERPRISE ARCHITECTURE DIAGRAM (TEXTUAL STRUCTURE)

Below is the official logical architecture representation that can be converted into a formal diagram for presentation and procurement alignment.

6.1 High-Level Architecture Flow

```
[CITIZENS / BUSINESSES]
          ↓
[Presentation Layer]
  - Service Counters
  - Queue Management
  - Citizen Portal (Future)
          ↓
[Application / Workflow Layer]
  - Service Engine (47 Services)
  - SLA Manager
  - Document Management
  - Notification Engine
          ↓
[API Gateway / Integration Layer]
  - Authentication Middleware
  - API Registry
  - Traffic Controller
  - Logging & Monitoring
          ↓
-----------------------------------------
|         Bureau Backend Systems        |
```

```
|---------------------------------------|
| Land Registry System                  |
| Revenue Tax System                    |
| Trade Licensing System                |
| Investment Registry                   |
| Health & Education Systems            |
| New Backend Systems (Category B)      |
-----------------------------------------
          ↓
[Data Layer]
  - Transaction DB
  - Payment Ledger
  - Document Storage
  - Master Data Registry
  - Analytics Warehouse
          ↓
[Infrastructure Layer]
  - Hosting Servers
  - Network Core
  - Storage Arrays
  - Firewall
  - Backup Systems
          ↓
[National MESOB API Mesh]
```

6.2 Integration Flow Example – Lease Transfer

Citizen → MESOB Counter
→ Workflow Engine
→ API Gateway
→ Land Registry API
→ Revenue API
→ Payment Confirmation
→ Certificate Issuance
→ Dashboard Update

All steps logged.

---

7. HOSTING TOPOLOGY OPTIONS

Hosting must support:

- Reliability
- Security
- Growth
- Cost control

Three models exist.

7.1 Model 1 – Fully Managed Ethio Telecom Hosting

Architecture:

MESOB application + databases hosted in telecom cloud/data center.

Advantages:

- Quick deployment
- Integrated connectivity
- Single vendor responsibility

Risks:

- Vendor lock-in
- Cost escalation
- Migration complexity

Contract must guarantee:

- Data export rights
- Source code ownership clarity
- SLA penalty clauses

7.2 Model 2 – Hybrid Hosting (Recommended)

Structure:

Core application hosted by telecom
Critical databases replicated locally
Backup stored independently

Advantages:

- Data sovereignty
- Disaster resilience
- Negotiation leverage

Requires:

- Strong governance
- Clear integration architecture

7.3 Model 3 – Fully Local City Data Center

Advantages:

- Full control
- Maximum autonomy

Risks:

- High capital expenditure
- High maintenance burden
- Skilled staff requirement

Not recommended for Phase 1 unless strong IT capacity exists.

---

8. LOCALIZED DATA CENTER REQUIREMENTS

Even if hosting is external, MESOB building must include:

8.1 On-Premise Infrastructure Room

Minimum requirements:

- Secured server rack area
- Temperature control
- Fire suppression system
- Restricted access
- CCTV monitoring

8.2 Core Hardware Requirements (Logical)

- Edge firewall appliance
- Core network switch
- Backup storage unit
- Local application cache server (optional)
- UPS system (minimum 2 hours backup)
- Generator integration

8.3 Network Redundancy

Must include:

- Primary fiber link
- Secondary backup link
- Automatic failover configuration

---

9. SECURITY ZONING ARCHITECTURE

Security must be segmented into zones.

9.1 Security Zones

Zone 1 – Public Access Zone

- Service counters
- Queue kiosks

Zone 2 – Application Zone

- Workflow servers
- API gateway

Zone 3 – Database Zone

- Transaction DB
- Payment Ledger
- Document Storage

Zone 4 – Management Zone

- Admin consoles
- Monitoring systems

Zone 5 – External Integration Zone

- National API Gateway
- Bank payment integration

Each zone must be separated by firewall rules.

9.2 Zero-Trust Principle

No internal system automatically trusted.

Each API call must:

- Authenticate
- Authorize
- Log

9.3 Security Monitoring

Required:

- Intrusion detection
- Traffic anomaly detection
- API abuse monitoring
- Payment fraud alerts

---

10. SCALABILITY & FUTURE-PROOFING MODEL

MESOB must scale beyond Phase 1.

10.1 Service Expansion Capacity

Architecture must support:

- 100+ services (Phase 2 & beyond)
- Online self-service portal
- Mobile integration
- Smart city integration

10.2 Horizontal Scaling

Application layer must support:

- Load balancing
- Multi-instance deployment
- Elastic resource scaling

10.3 Data Growth Projection

Estimated:

- 3–5 TB growth every 3–5 years
- Document archive expansion
- Analytics dataset growth

Storage must be modular.

10.4 Future Integrations

Potential future connections:

- Regional government systems
- National ID platform
- Financial institutions
- GIS urban planning system
- Smart inspection tools

Architecture must not require full redesign for integration.

ENTERPRISE ARCHITECTURE RISK WARNINGS

If architecture is not properly structured:

- API chaos will occur
- Vendor-specific lock-in will trap city
- Phase 2 expansion becomes impossible
- SLA monitoring will collapse
- Security breaches become likely

Enterprise discipline is non-negotiable.

DOCUMENT 6 COMPLETE

This concludes:

✓ Enterprise architecture diagram structure
✓ Hosting topology models
✓ Local data center requirements
✓ Security zoning architecture
✓ Scalability & future-proofing strategy

Reply with:

“Proceed with Document 7, Part 1.”

Next we begin:

Physical Infrastructure & MESOB Building Setup
(Room planning, service windows, manpower allocation, hardware placement, and service capacity modeling).